A Multidimensional Feature-Driven Ensemble Model for Accurate Classification of Complex Power Quality Disturbance

Yulong Liu, Ding Yuan, Hongwei Fan, Tao Jin, *Senior Member, IEEE*, and Mohamed A. Mohamed, *Senior Member, IEEE*

***Abstract*— As the proportion of power electronics-related facilities in modern power systems increases, the types of power quality disturbances (PQDs) tend to become more complex. Traditional methods struggle to accurately perform the classification task of complex PQDs under artificial empirical guidance. This article proposes a multidimensional feature-driven ensemble model for the accurate classification of complex PQDs, which can complete the self-learning function from data. Unlike existing deep learning-based methods, this model considers both spatial features in the time–frequency domain and temporal relational features. Based on fully convolutional networks (FCNs) and bidirectional gated recurrent unit (BiGRU), submodules suitable for multidimensional features mining are constructed separately. Meanwhile, the squeeze-and-excitation network (SENet) is introduced to complete the computation of the channel attention mechanism for each convolutional layer, which effectively improves the training efficiency and classification accuracy of the model. The proposed model has been thoroughly tested to validate its effectiveness on a synthetic dataset consisting of 71 different types of PQDs under varying signal-to-noise ratios (SNRs). Additionally, the model has been proven to be robust in the face of external factors, such as dc offset, frequency variations, and phase jumps. To further demonstrate its reliability, the proposed model has been tested on real PQDs generated from an ac power source. Results from both simulation and experimentation have conclusively shown that the proposed method is superior to existing deep learning-based methods for the classification of complex PQDs.**

***Index Terms*— Bidirectional gated recurrent unit (BiGRU), classification, complex power quality disturbances (PQDs), deep learning, fully convolutional network (FCN).**

I. Introduction

RENEWABLE energy development is the main solution to energy shortages and environmental pollution. Gradually increasing the proportion of renewable energy generation (REG), especially photovoltaic and wind power, is the consensus view of most countries [1]. Moreover, the universal adoption of electric vehicles (EVs) is also an effective way to address the above two issues [2]. However, these changes enrich the power system with power electronics, which inevitably affects the power quality on the supply side. If nothing is done to control these hazards, they will not only affect the power consumption experience of end users but also endanger the operational stability of the grid [3]. The effects of these facilities will initially appear as power quality disturbances (PQDs). If these PQDs can be detected accurately and promptly, action can be taken to prevent further serious power quality mass issues [4].

Addressing the classification of complex PQDs is a hot topic of current research. Scholars have delved into two main directions, which are traditional methods that consist of signal analysis algorithms and classifiers and deep learning-based methods that enable self-learning of the data. For traditional methods, signal processing methods, such as wavelet transform (WT) [5], Stockwell transform (ST) [6], empirical mode decomposition (EMD) [7], and variational mode decomposition (VMD) [8], are first used to obtain the reconstructed components of the signal in different information. Then, after artificial feature selection, a classification model is designed in combination with artificial intelligence techniques, such as decision tree (DT) [9], support vector machine (SVM) [7], k-Nearest Neighbor (KNN) [10], extreme learning machine (ELM), and so on, which can establish mapping relations from continuous features to discrete labels and finally achieve the classification of PQDs. There are two areas of focus in the study of such methods. One is to improve the accuracy and efficiency of signal processing methods, such as the proposal of fast ST [11], double resolution ST [12], and tunable-Q WT [13]. The remaining one is to enrich the selection range of artificial features, such as time-dependent spectral features [14] and multiscale fluctuation dispersion entropy [4]. It is known from the existing literature that traditional methods require manual feature selection and there is no uniform evaluation criterion to objectively guide this crucial step, which can easily result in redundant or meaningless features. Moreover, when the types of PQDs increase, it is difficult for manual experience to satisfy both accuracy and noise robustness.

Deep learning-based classification methods, which rely less on specific mathematical models and are good at feature self-learning from the dataset, are able to solve the aforementioned

Manuscript received 16 January 2023; revised 21 March 2023; accepted 22 March 2023. Date of publication 10 April 2023; date of current version 27 April 2023. This work was supported in part by the Chinese National Natural Science Foundation under Grant 51977039 and in part by the Central Government Guiding Local Science and Technology Development Project under Grant 2021L3005. The Associate Editor coordinating the review process was Dr. Alessandra Galli. *(Corresponding authors: Tao Jin; Mohamed A. Mohamed.)*

Yulong Liu, Ding Yuan, Hongwei Fan, and Tao Jin are with the College of Electrical Engineering and Automation, Fuzhou University, Fuzhou 350116, China (e-mail: n190110008@fzu.edu.cn; n190110008@fzu.edu.cn; 200120001@fzu.edu.cn; jintly@fzu.edu.cn).

Mohamed A. Mohamed is with the Electrical Engineering Department, Faculty of Engineering, Minia University, Minia 61519, Egypt (e-mail: dr.mohamed.abdelaziz@mu.edu.eg).

Digital Object Identifier 10.1109/TIM.2023.3265756

1557-9662 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 02:08:05 UTC from IEEE Xplore. Restrictions apply.

problems well. Thus, they have become the dominant trend in the classification of PQDs. There are two main directions for deep learning-based methods. One is to convert the 1-D sampled sequences of PQDs into 2-D images through visualization techniques and combine them with deep learning-based image classification models to extract features to complete the classification task. In [15], the sampled sequence is converted into a 2-D grayscale image, while the features are amplified with the help of image enhancement techniques. To further highlight the local characteristics, Wigner–Ville distribution [16] and Gramian angular summation field [17] are introduced to visualize the PQD signal. Visualized PQDs can be easily identified by deep learning-based image classification methods. The key point of such methods is how to maximize the features of PQDs in the 2-D view. Another direction is to take the 1-D sampled sequence as the model input and apply the spatial feature extraction capability of convolutional neural networks (CNNs) and temporal feature memory capability of recurrent neural network (RNN) to perform the classification of PQDs. From the perspective of spatial features, Wang et al. [18] proposed a deep CNN (DCNN) stacked with multiple units to deal with 1-D sequences, which can automatically extract disturbed high-dimensional features and achieve accurate classification. To further improve the accuracy and efficiency of PQDs classification, Gong and Ruan [19] introduced an improved DCNN, whose feature extraction ability relies on 1-D-modified inception-residual network (1-D-MIR) modules. Simulation results show that the proposed model can guarantee both a higher convergence rate and stronger generalization ability. From the perspective of PQDs temporal relationship, a simple gated recurrent network (SGRN) based on RNN was proposed in Zu and Wei [20]. Compared to RNN, long short-term memory (LSTM) and gate recurrent unit (GRU) involve fewer parameters, which reduces the amount of computation while ensuring accuracy. Comparing the above two directions, classification methods using 2-D images as an input not only complicate the subsequent network structure and the dataset but also easily lead to feature loss. Furthermore, as the variety of PQDs increases, it becomes difficult to express distinct features on images. In contrast, directly feeding 1-D sampled sequences into the deep learning network can mine features more efficiently.

Nowadays, deep learning is rapidly developing, which is more favorable for the accurate classification of complex PQDs. However, there are three shortcomings in the existing-related studies as follows. First, there are too few types of complex PQDs under consideration, and as the number of disturbance sources increases, it becomes difficult to meet the requirements for the identification of complex PQDs. Second, noise interference is unavoidable in practical applications. With the increasing types of PQDs, it is difficult for the existing methods to guarantee both accuracy and noise robustness. Third, most deep learning-based methods singularly focus on time-domain spatial features or temporal relational features, leading to a limited range of representation learning, which makes it difficult to further improve the classification accuracy.

To meet the application requirements of complex PQDs classification, a multidimensional feature-driven ensemble model is proposed in this article. The proposed method considers both spatial features in the time–frequency domain and temporal relational features, for which corresponding submodules are constructed. Finally, the classification task of PQDs is completed by fusing the features of each submodule. The dataset used contains a total of 71 types of PQDs, covering all possible single, double, triple, and quadruple PQDs classified according to IEEE STD 1159-2019. Both simulations and experiments are conducted to validate the proposed model, whose results are compared with other mainstream deep learning-based methods and traditional methods. Considering the existing work, the main contributions of this article can be listed as follows.

1) Instead of representation learning in a single feature domain, the proposed method integrates spatial features in the time–frequency domain and temporal relational features through an ensemble model, which not only achieves high recognition accuracy for 71 PQDs but also shows strong robustness in the noise environment.
2) For the parallel fully convolutional network (FCN) that extracts spatial features in the time–frequency domain, the channel attention mechanism of squeeze-and-excitation network (SENet) is introduced, which can automatically identify the importance of each convolutional channel and effectively improve the generalization ability of the model.
3) For the submodule of bidirectional gated recurrent unit (BiGRU) that extracts temporal relational features, the input sequence is cyclically segmented according to the signal characteristics of PQDs, which alleviates the gradient vanishing problem and improves temporal memory capability.

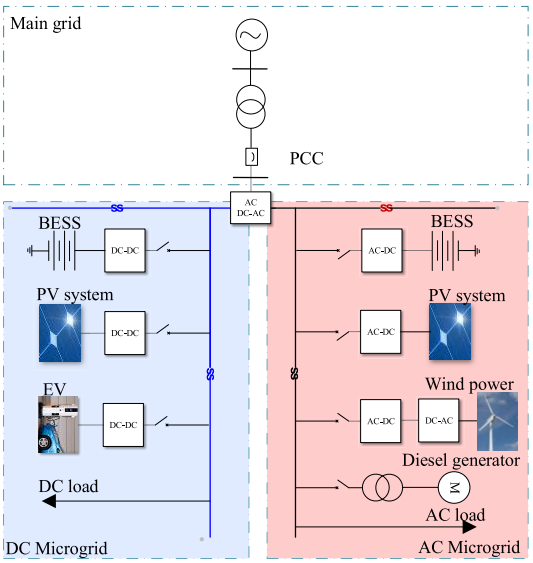


Fig. 1. Distribution network integrated with ac/dc microgrids.

II. Complex PQDs and the Proposed Ensemble Model

A. Complex PQDs

As shown in Fig. 1, the distribution network integrates REG, EV, battery energy storage system (BESS), and so on through

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 02:08:05 UTC from IEEE Xplore. Restrictions apply.

TABLE I
TYPE OF PQDS GENERATED BY REG AND EV OPERATIONS

Facilities	Related Operations	Types of PQDs
Photovoltaics	Synchronization, outage, island, sudden changes in solar insolation	Swell, sag, flicker, harmonics, impulsive, notching
Wind power	Synchronization, outage, island, sudden changes in wind speed	Swell, sag, flicker, oscillatory, impulsive, harmonics, fluctuations
EV	Fast charge, disconnection	Swell, sag, harmonic

microgrids, which is a trend in the future [21]. With the help of microgrids, energy shortages and environmental pollution can be alleviated. However, PQDs due to power electronics and natural resource instabilities cannot be neglected. The types of PQDs produced by these facilities change as the operations and connections change. As the energy demand grows, the number of these facilities will expand, which will considerably increase the probability of the superposition of different PQDs. Therefore, PQDs that need to be considered are no longer the conventional single ones, but complex ones.

Many scholars have paid attention to the PQDs caused by these facilities at the point of common coupling (PCC) [22]. The PQDs generated by wind power and photovoltaics mainly consist of the following two points. One is the effect of resource fluctuations and discontinuities. When large amounts of volatile wind power and photovoltaics are integrated, the grid is affected by surge current, which causes voltage swell, sag, flicker, and so on. The other is the effect of harmonics. Wind power, for example, is inherently equipped with power electronics that can cause harmonic problems. The constant velocity generators in the soft-start phase will generate certain harmonics, while for variable-speed generators, more serious harmonics can be generated if the power electronics are in the range of the generated harmonics. By reviewing the relevant literature [4], [8], [22], the PQDs that could be generated by REG and EV are summarized and listed in Table I. Furthermore, Xu et al. [8] pointed out that when these facilities operate simultaneously, PQDs can be severed and even new types of PQDs can be generated.

Synthetic data for the single PQDs mentioned above can be generated using IEEE STD 1159-2019 and mathematical equations, and for complex PQDs, the essence is the combination of single PQDs, which can be mathematically formulated to form synthetic data through the following equation:

$$x = \mathrm{Mul}(t) \cdot \sin(\omega t) + \mathrm{Add}(t) \tag{1}$$

where t is the time series based on the sampling frequency, ω is the angular frequency under the fundamental frequency, and $\mathrm{Mul}(t)$ is the PQD in the form of multiplication, such as voltage sag, swell, and interruption. $\mathrm{Add}(t)$ is the PQD in the form of addition, such as harmonic, oscillatory, and noise. Some single PQDs and complex PQDs are selected to display their waveforms, as shown in Fig. 2.

Fig. 2(a) illustrates the single PQD of "harmonics" generated by using (2) and containing the third, fifth, and seventh harmonics. The harmonic distortion is present throughout, and the amplitude (represented by ϑ_o) of each component is 0.08, 0.12, and 0.11 p.u., respectively.

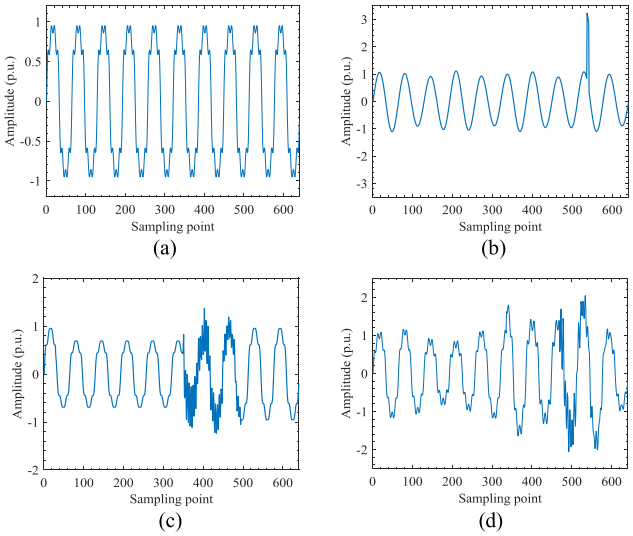


Fig. 2. Some examples of PQDs waveforms. (a) Harmonics. (b) Fluctuations + impulsive. (c) Sag + oscillatory + harmonics. (d) Swell + oscillatory + harmonics + fluctuations.

$$x_1 = \sin(\omega t) + \sum_{o=3,5,7} \vartheta_o \sin(o\omega t). \tag{2}$$

The double PQD of "fluctuations + impulsive" in Fig. 2(b) is generated through (3). Both types of disturbance are combined with the standard signal by multiplication. The amplitude g_f and frequency μ of the fluctuation are 0.05 p.u. and 16 Hz, respectively. The amplitude g_I and duration $(t_2 - t_1)$ of the impulsive are 3 p.u. and 1.5 ms. s_f ( ) is a step function.

$$x_2 = [1 + g_I(s_f(t - t_1) - s_f(t - t_2))] \times [1 + g_f \sin(2\pi\mu t)] \sin(\omega t). \tag{3}$$

The triple PQD labeled "sag + oscillatory + harmonics" shown in Fig. 2(c) is generated by using (4). The sag component has an amplitude g_s of 0.3 p.u. and a duration $(t_2 - t_1)$ of 100 ms. The oscillation component has a frequency f_o of 850 Hz, an amplitude g_o of 0.6 p.u., a time reference κ of 35 ms, and a duration $(t_4 - t_3)$ of 45 ms. Additionally, the amplitudes ϑ_o of the third, fifth, and seventh harmonic components are 0.12, 0.10, and 0.05 p.u., respectively

$$x_3 = \{1 - g_s[u(t - t_1) - u(t - t_2)]\} \sin(\omega t) + \sum_{o=3,5,7} \vartheta_o \sin(o\omega t) + g_o e^{\frac{-(t-t_3)}{\kappa}} \times [u(t - t_3) - u(t - t_4)] \sin(2\pi f_o(t - t_3)). \tag{4}$$

In Fig. 2(d), the quadruple PQD of "swell + oscillatory + harmonics + fluctuations" is generated by (5). The swell has an amplitude g_{sw} of 0.5 p.u. and lasts for 100 ms, while the oscillatory has a frequency f_o of 600 Hz, an amplitude g_o of 0.5 p.u., a time reference κ of 20 ms, and a duration $(t_4 - t_3)$ of 30 ms. The third, fifth, and seventh harmonics have amplitudes ϑ_o of 0.09, 0.10, and 0.10 p.u., respectively. Additionally, the fluctuation has an amplitude g_f

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 02:08:05 UTC from IEEE Xplore. Restrictions apply.

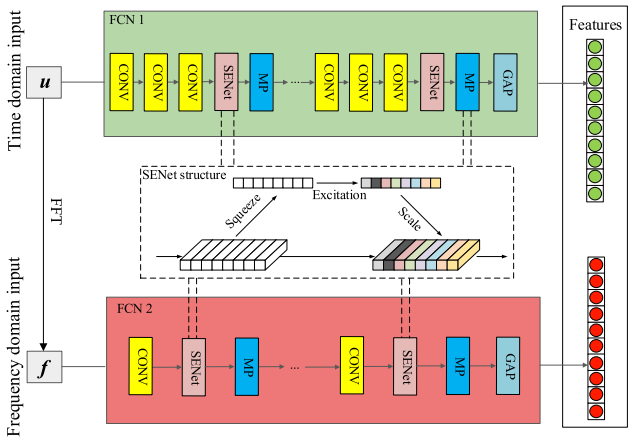


Fig. 3. Detailed composition of the parallel FCN.

of 0.04 p.u. and a frequency μ of 10 Hz.

$$x_4 = \{1 + g_{sw}[u(t-t_1) - u(t-t_2)]\}[1 + g_f \sin(2\pi\mu t)] \times \sin(\omega t) + \sum_{o=3,5,7} \vartheta_o \sin(o\omega t) + g_o e^{\frac{-(t-t_3)}{\kappa}} \times [u(t-t_3) - u(t-t_4)] \sin(2\pi f_o(t-t_3)). \quad (5)$$

B. Parallel FCN Optimized by SENet

Convolutional networks are most commonly used to extract spatial features, as they can acquire local and abstract features through shallow and deep convolutional layers, respectively. FCN is a special network that belongs to CNN. Compared to regular CNN, FCN omits fully connected layers at the end, which results in the overall features being returned instead of the classification result. It is simpler and more efficient to extract spatial features using the FCN containing only convolution and pooling layers.

First, a parallel FCN with two-channel input is constructed for time–frequency domain sequences. The corresponding single-line depth is designed separately based on the complexity of the sequence. Second, all convolution kernels in the model are tuned to be 1-D to fit the format of the input sequence. Finally, the global average pooling (GAP) layer is applied after the last convolution layer to refine the global features and reduce the model computation. Compared with the global max pooling (GMP) layer, GAP pays more attention to the overall information, making the model more robust under interference conditions. In addition, to improve the training efficiency and accuracy of the model, the channel attention mechanism is further considered, that is, embedded SENet after each convolution operation. SENet enhances important features to improve accuracy by modeling the correlation between feature channels. The composition of the proposed parallel FCN is shown in Fig. 3.

The input layer of the two-channel FCN consists of $\boldsymbol{u}$ and $\boldsymbol{f}$, which represent the time–frequency domain sequences of PQDs. Building upon discrete Fourier transform (DFT), fast Fourier transformation (FFT) speeds up the frequency spectrum calculation process through decomposition and recursion, making it a preferred tool for obtaining frequency spectrum input $\boldsymbol{f}$ from the voltage sampled signal $\boldsymbol{u}$.

$$\begin{cases} u = [u_1, u_2, \ldots, u_n] \\ f = [f_1, f_2, \ldots, f_{n/2}] \end{cases} \quad (6)$$

where the total number of sequences in the time and frequency domain is n, $n/2$, respectively. When the output from the $l-1$ layer passes through the convolution layer, the moving convolution kernel with length k performs a convolution operation on each element in $\boldsymbol{u}^{(l-1)}$ and $\boldsymbol{f}^{(l-1)}$ through the nonlinear activation function, and the outputs $\boldsymbol{u}_{\text{conv}}^{(l)}$ and $\boldsymbol{f}_{\text{conv}}^{(l)}$ are as follows:

$$\begin{cases} u_{\text{conv}}^{(l)} = \sigma\left(\sum_{i \in n-k+1} W_u^{(l)} \cdot u_i^{(l-1)} + B_u^{(l)}\right) \\ f_{\text{conv}}^{(l)} = \sigma\left(\sum_{i \in m-k+1} W_f^{(l)} \cdot f_j^{(l-1)} + B_f^{(l)}\right) \end{cases} \quad (7)$$

where $\boldsymbol{W}^{(l)}$ and $B^{(l)}$ are the weights and bias of the convolution kernel corresponding to the lth layer, respectively. $\sigma(\cdot)$ represents the activation function, which is ReLU used in this article, and its calculation can be represented as follows:

$$\text{ReLU}(x) = \max(0, x). \quad (8)$$

After the convolution operation, the weights of each channel are computed through SENet to assign valuable channels to higher weights. Taking $\boldsymbol{u}_{\text{conv}}^{(l)}$ as an example, its dimension is $(1 \times m \times C)$. The squeezing operation is first performed by computing the global average of m elements of each channel to obtain the eigenmatrix $\mathbf{U}_{sq}^{(l)} = [u_{sq}^{(l)}, 1, \ldots, u_{sq}^{(l)}, c, \ldots, u_{sq}^{(l)}, C]$.

$$u_{\text{sq},c}^{(l)} = \frac{1}{m} \sum_{i=1}^{m} u_{\text{conv},c}^{(l)}(i). \quad (9)$$

Next, two fully connected layers are used to achieve dimensionality reduction and augmentation to compute the weights of the C-channels $\boldsymbol{s}^{(l)} = [s_1^{(l)}, \ldots, s_c^{(l)}, \ldots, s_C^{(l)}]$, and through the following equation, the gate mechanism can be defined:

$$s^{(l)} = \sigma\left(g\left(\mathbf{U}_{\text{sq}}^{(l)}, \mathbf{W}\right)\right) = \sigma\left(\mathbf{W}_2 \delta\left(\mathbf{W}_1 \mathbf{U}_{\text{sq}}^{(l)}\right)v\right) \times \mathbf{W}_1 \in \mathbf{R}^{\frac{C}{r} \times C}, \quad \mathbf{W}_2 \in \mathbf{R}^{C \times \frac{C}{r}} \quad (10)$$

where the activation functions σ and δ are sigmoid and ReLU, respectively. $\mathbf{W}_1$ and $\mathbf{W}_2$ are the parameters of the dimension reduction and augmentation layers with dimension reduction ratio r. Finally, the different channels are multiplied by their corresponding weights to implement the attention mechanism for the key channels.

$$\boldsymbol{u}_{\text{se},c}^{(l)} = s_c^{(l)} \cdot \boldsymbol{u}_{\text{conv},c}^{(l)}. \quad (11)$$

In the pooling layer, spatial merging via max pooling (MP) windows can reduce the feature size while maintaining information. As the inputs in both the temporal and frequency domains are 1-D sequences, the MP operation is carried out using 1-D calculations. The pooling window moves from left to right with a fixed stride length, and the maximum value within the window is computed until the entire sequence is

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 02:08:05 UTC from IEEE Xplore. Restrictions apply.

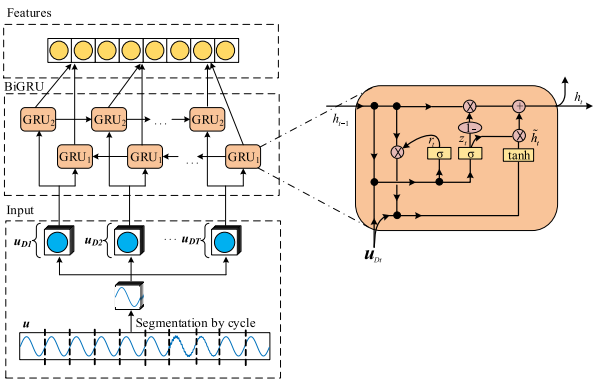


Fig. 4. Composition of the BiGRU submodule.

covered. If the length of the pooling window is p, the output $\boldsymbol{u}_{mp,c}^{(l)}$ is as follows:

$$\boldsymbol{u}_{\mathrm{mp},c}^{(l)}(i) = \max(\boldsymbol{u}_{\mathrm{se},c}^{(l)}(i), \boldsymbol{u}_{\mathrm{se},c}^{(l)}(i+1), \ldots, \boldsymbol{u}_{\mathrm{se},c}^{(l)}(i+p-1)), \quad (i = 1, \ldots, m-p+1). \tag{12}$$

The GAP is used in the final pooling in the FCN submodule to obtain the global spatial feature, which consists of the global average information of all convolution kernels. The process of GAP is to compute the average value of each feature channel and combine them to form $\boldsymbol{u}_{\mathrm{GAP}}$.

C. BiGRU Optimized by Sequence Reshaping

Compared to traditional RNN and LSTM, BiGRU is a more lightweight, efficient, and memory-capable option, making it the ideal choice for extracting temporal relationship features of PQDs. Additionally, to mitigate computational complexity and the gradient vanishing problem, the input sequence is periodically reshaped.

The composition of the BiGRU submodule and the treatment of the data are shown in Fig. 4. First, the input signal $\boldsymbol{u}$ is divided into T group vectors according to the cycle to form the reshaped data $[\boldsymbol{u}_{D1}, \ldots, \boldsymbol{u}_{Dt}, \ldots, \boldsymbol{u}_{\mathrm{DT}}]$. The input is then fed to a BiGRU network composed of forward and backward GRU cells in parallel. Finally, groups of forward and backward temporal correlated eigenvalues are generated and merged.

The tth GRU cell obtains the state of the reset gate r_t and update gate z_t via the state h_{t-1} transmitted from the previous cell and the current input $\boldsymbol{u}_{Dt}$. The two gate vectors are updated by the following equation:

$$\begin{cases} z_t = \sigma(\mathbf{W}_z \cdot [h_{t-1}, \boldsymbol{u}_{Dt}]) \\ r_t = \sigma(\mathbf{W}_r \cdot [h_{t-1}, \boldsymbol{u}_{Dt}]) \end{cases} \tag{13}$$

where $\mathbf{W}_z$ and $\mathbf{W}_r$ are the weight matrices of the gate z_t and r_t. σ is the activation function of the sigmoid. The candidate activation vector and output vector can be updated as follows:

$$\begin{cases} \tilde{h}_t = \tanh(\mathbf{W} \cdot [r_t * h_{t-1}, \boldsymbol{u}_{Dt}]) \\ h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t \end{cases} \tag{14}$$

where $\mathbf{W}$ is the weight matrice of the state $\tilde{h}_t$, and the hidden state h_t will be transmitted to the next moment.

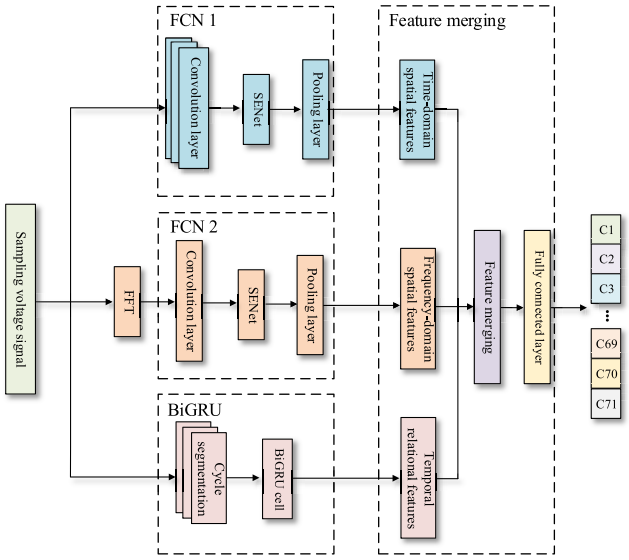


Fig. 5. Framework of the proposed ensemble model.

D. Overall Framework of the Proposed Ensemble Model

The proposed ensemble model extracts spatial features in the time–frequency domain and temporal relational features through the modules introduced above, respectively. Subsequently, these features are integrated in a serial manner through the concatenate layer. Finally, all features are normalized and classified by the dense layer. The dense layer performs a nonlinear calculation of the input through the activation function of Softmax to complete the classification task. For example, the probability $P(X_d)$ that the input sample X belongs to class d can be calculated as follows:

$$P(X_d) = \mathrm{Softmax}(X_d) = \frac{\exp(X_d)}{\sum_{k=1}^{K} \exp(X_k)} \tag{15}$$

where K is the total number of PQDs classes. X_d is the output value of the fully connected layer. The dth class corresponding to the maximum $P(X_d)$ is the predicted PQDs class.

The overall framework of the proposed ensemble model is illustrated in Fig. 5, which contains the FCN module, the BiGRU module, and the feature mergence module. Two FCN modules extract spatial features in the time–frequency domain through convolution, channel attention mechanism, and pooling. At the same time, the signal is delivered to the BiGRU module for temporal relational feature extraction. The final classification task is done in the feature mergence module.

III. Dataset and Model Building

A synthetic dataset with 71 classes of PQDs is generated through MATLAB for training and testing the model, and the classification model is built in a python environment.

A. Details of Dataset and Model

The eight single PQDs are classified into four categories by IEEE STD 1159-2019, which are root-mean-square variations, transients, waveform distortion, and fluctuations. Based on all possible combinations of the above four categories,

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 02:08:05 UTC from IEEE Xplore. Restrictions apply.

TABLE II
HARDWARE MODELS AND SOFTWARE VERSIONS

Hardware	Model	Software	Version
CPU	Intel Core i9-9820X @3.3.0GHz	Python	3.6
		Keras	2.3.1
GPU	NVIDIA GeForce RTX 2080 8G	TensorFlow-gpu	2.0.0
RAM	DDR4 32GB	CUDA	10.0
HDD	SSD 1TB	cuDNN	7.6.5

23 classes of double PQDs, 28 classes of triple PQDs, and 12 classes of quadruple PQDs are considered in the synthetic dataset [23]. The parameters involved in each disturbance are randomly set according to IEEE STD 1159-2019, such as duration, amplitude, frequency, and so on. Since all samples are generated through mathematical formulas, it is possible to establish a balanced dataset for model training. However, data imbalance may occur in practical applications. To address this, synthetic data based on mathematical formulas can be used to supplement the dataset and alleviate this problem.

There are 10 000 data for each class, half of which are noise-free and the remaining half subject to 20-dB white noise, and the dataset is split into training and validation sets with a ratio of 8:2. According to the parameter range in the standard and the settings in the existing literature [4], [15], [18], the dataset is sampled at a frequency of 3200 Hz with a 10-cycle length. Higher sampling frequencies can result in increased processing and storage costs, as well as introduce additional noise and distortion.

The model is trained using cross-validation, where the optimal one is determined based on the classification accuracy of the validation set. The Keras framework is applied to build the model and complete the classification task of PQDs. The configurations and versions of the relevant hardware and software are shown in Table II.

Table III gives the network parameters and setup of the proposed model. f and k are the number of filters and the size of the convolution kernel, respectively. s represents the stride of the filter, and r is the reduction ratio of SENet. Different initial learning rates are tried during training, which affects the training efficiency and generalization ability as known from the trends of the training curves. The model is trained optimally when 0.001 is the initial learning rate, which is set as the initial learning rate. Also, strategies for learning rate decay and early stopping are introduced.

Choosing an appropriate batch size and optimizer can help improve the training efficiency and accuracy of the model. The optimal configuration is determined by comparing the performance of five commonly used optimizers with different batch sizes, the results of which are shown in Fig. 6. The average accuracy of the Adam-based results is 96.67%, which is ahead of the other four optimizers, and thus, Adam is identified as the training optimizer. For batch sizes below 64, the classification accuracy increases with batch size, but for batch sizes above 64, the results are inversely proportional. Therefore, 64 is set as the batch size for network training.

TABLE III
PARAMETERS AND SETUP OF THE PROPOSED ENSEMBLE MODEL

Block	Layer	Parameters	Activation function
FCN1	Convolution_1 *3	f=16, k=3, s=1	ReLU
	SENet_1	units=16, r=16	ReLU+Sigmoid
	MP_1	p=3, s=1	none
	Convolution_2 *3	f=32, k=3, s=1	ReLU
	SENet_2	units=32, r=16	ReLU+Sigmoid
	MP_2	p=3, s=1	none
	Convolution_3 *3	f=64, k=3, s=1	ReLU
	SENet_3	units=64, r=16	ReLU+Sigmoid
	GAP_1	none	none
FCN2	Convolution_4	f=16, k=3, s=1	ReLU
	SENet_4	units=16, r=16	ReLU+Sigmoid
	MP_4	p=3, s=1	none
	Convolution_5	f=32, k=3, s=1	ReLU
	SENet_5	units=32, r=16	ReLU+Sigmoid
	MP_5	p=3, s=1	none
	Convolution_6	f=64, k=3, s=1	ReLU
	SENet_6	units=64, r=16	ReLU+Sigmoid
	GAP_2	none	none
BiGRU	BiGRU	units=128	none
Feature Mergence	Concatenate	none	none
	Dense_1	units=256	none
	Batch Normalization	none	none
	Dense_2	units=128	none
	Dense_3	units=72	Softmax

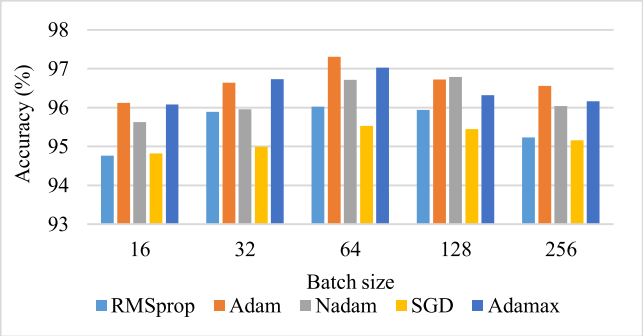


Fig. 6. Training results for various batch sizes with different optimizers.

B. Model Training Process

To demonstrate the training process of the proposed model, the accuracy and loss are recorded for each epoch of the training and validation sets, which are derived at both 20-dB noise and noise-free cases. To demonstrate the benefits of merging features and optimization strategies for the PQD classification model, a convolutional network with only time-domain spatial features is set up as the comparison model. Without the feature merging module, the fully connected layer is directly connected to the global pooling layer, resulting in the comparison model being a typical CNN, and the parameter settings of the CNN are consistent with the proposed model. The process records for the two models are shown in Fig. 7.

First, the two results are compared in terms of training efficiency. The proposed model converges in 27th epoch, which is 17 epochs ahead of the single-feature CNN model. Second,

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 02:08:05 UTC from IEEE Xplore. Restrictions apply.

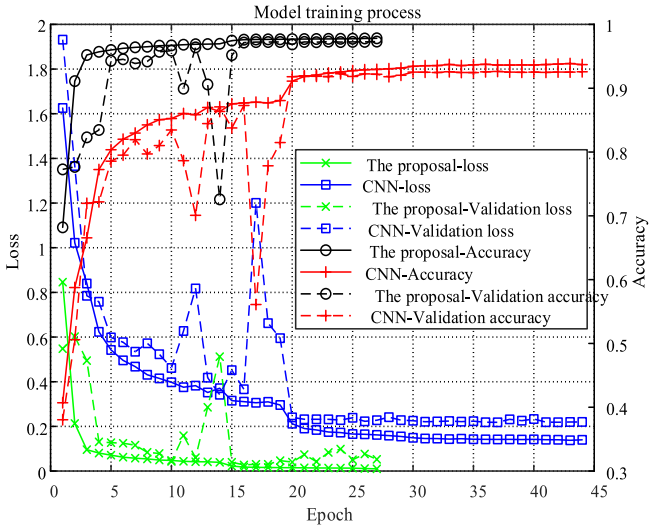


Fig. 7. Model training process.

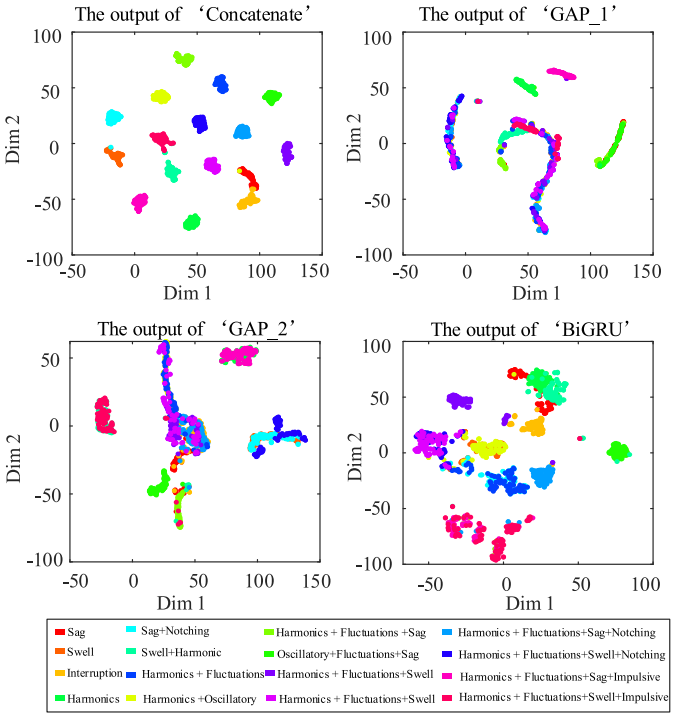


Fig. 8. Visualization of the feature extraction capability.

for the proposed model, the final accuracy gap between the training and validation set is only 0.51%, which is 0.61% less than that of the CNN model. Thus, the proposed model is less overfitting. Meanwhile, the classification accuracy of the proposed model has a clear advantage, which is 4.73% higher than the CNN model for the validation set. From the overall results, the proposed optimization strategy effectively improves the training efficiency and classification accuracy of the model.

IV. Results and Discussion

The effectiveness of the proposed method is demonstrated with synthetic and real signals. Meanwhile, the performance of the proposed model is compared with the state-of-the-art deep learning-based methods and traditional methods.

A. Visualization of the Feature Extraction Capability

With the help of t-distributed stochastic neighbor embedding (T-SNE), the feature extraction capability can be visualized by projecting the output onto a 2-D embedding space. To show the superiority of the ensemble strategy, the proposed model is compared with three submodules in terms of visualization results. Four types of single PQDs, double PQDs, triple PQDs, and quadruple PQDs are randomly selected as input data, with 100 data per type. After each set of data is fed into the trained model, the outputs of "GAP_1," "GAP_2," "BiGRU," and "concatenate" are recorded separately for visual computation and set "exact" and "euclidean" as T-SNE's projection algorithm and distance calculation method. The visualization results are shown in Fig. 8.

It is clear from Fig. 8 that the visualization results for the output of the feature by the three submodules have different levels of overlap issues. Distributions representing spatial features in the frequency domain overlap the most due to differences in the features concentrated at the main frequency points. In contrast, the visual distribution of the ensemble model is able to distinguish different PQDs well, which demonstrates the effectiveness of the proposed model to merge the three types of features.

TABLE IV
Classification Results in Different Noise Environments

Models	Classification results in different noise environments (Accuracy% ± Standard deviation)			
	20 dB	30 dB	40 dB	Noise-free
DCNN	91.88±3.52	95.13±1.72	95.07±1.44	97.91±0.83
SGRN	85.72±3.12	88.90±2.39	93.41±2.69	93.49±2.02
1D-MIR	90.55±2.47	92.51±2.07	94.08±1.24	97.80±1.19
Proposal	94.74±3.05	98.79±0.89	99.21±0.08	99.46±0.06

B. Model Testing in Different Noise Environments

Testing the model for noise robustness is a critical step for practical application. The test signals consist of voltage samplings affected by 20-, 30-, and 40-dB white noise, as well as a noise-free dataset for comparison. There are 2000 sets of data for each PQDs type in different noise environments. In addition, three state-of-the-art deep learning-based methods are set as comparison methods, namely, DCNN [18], SGRN [20], and 1-D-MIR [19]. The parameters and setup of the compared models are in accordance with the requirements listed in the relevant references, and the training environment and dataset are the same for all models. The classification results for each model are presented in Table IV.

Without noise interference, the proposed model achieves the highest classification accuracy of 99.46%, outperforming DCNN, SGRN, and 1-D-MIR by 1.55%, 5.97%, and 1.66%, respectively. The classification accuracy of all models suffers when the white noise intensity increases. Compared to the

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 02:08:05 UTC from IEEE Xplore. Restrictions apply.

other models, the proposed model shows strong robustness under interference from 30- and 40-dB white noise, with only a 0.25% and 0.67% decrease in accuracy. At the same time, it can be seen from the standard deviation that the results distribution of the proposed model is most concentrated. With the signal-to-noise ratio (SNR) low as 20 dB, the accuracy of the proposed model drops by 4.72%, and also known from the standard deviation that the accuracy of some PQDs types is seriously affected, but the accuracy of the proposal remains in the lead.

Specifically, Table V shows the classification accuracy of the proposed model for all PQDs under 20-dB white noise, as well as further analysis of its effectiveness. In Table V, the terms from left to right are the combinations of complex PQDs, and the numbers in parentheses denote the number of PQDs and their classification accuracy, respectively. As an example, the term marked in red means that the PQD of no. 61 is a quadruple PQD consisting of swell, impulsive, harmonic, and fluctuation, which is classified with 96.7% accuracy by the proposed model.

The analysis of the above table shows that the classification accuracy of some complex PQDs containing harmonics, oscillatory, notching, and fluctuations is greatly affected by 20-dB white noise. For example, the recognition accuracy of "sag + oscillatory + notching + fluctuations" is only 87.8%, which is caused by the weak feature expression of the distorted signal in the strong noise environment. However, overall, the average classification accuracy of the proposed model for single PQDs, double PQDs, triple PQDs, and quadruple PQDs can reach 96.0%, 94.6%, 94.9%, and 93.6% respectively, indicating that the proposed model can effectively distinguish complex PQDs in the presence of strong noise.

C. Robustness Testing for DC Offset, Frequency Variations, and Phase Jumps

In the integration of renewable energy sources into the power grid, it is necessary to consider the effects of dc offset, frequency variations, and phase jumps on the PQD classification model. In the following, the robustness of the proposed model will be tested one by one in these three interference environments. The first factor is the dc offset, which arises from the converter's inability to completely eliminate the dc component. According to IEEE STD 1159-2019, the typical voltage magnitude of dc offset ranges from 0% to 0.1%, and the typical duration is steady state. During testing, the magnitude of the dc offset is varied from 0% to 0.10% in steps of 0.01%, maintaining a steady state and accounting for the effects of white noise. In each noise environment, there are a total of 500 data for each PQD type under different levels of dc offset. The test results of the proposed model under the influence of dc offset in different noise environments are presented in Fig. 9, displayed in the form of average accuracy.

The impact of dc offset on the accuracy of the proposed model can be observed by the fluctuations of the surface along the x-axis in Fig. 9. Among the four noise environments represented on the y-axis, the proposed model is most affected

TABLE V
DETAILED CLASSIFICATION ACCURACY OF THE PROPOSED MODEL UNDER 20-dB WHITE NOISE

Single PQDs	Double PQDs	Triple PQDs	Quadruple PQDs
Normal (1, 95.1)	--	--	--
Swell (2,96.4)	Impulsive (10, 95.3)	Harmonics (33,97.9)	Fluctuations (61,96.7)
		Notching (34,97.3)	Fluctuations (62,97.4)
		Fluctuations (35,98.8)	--
	Oscillatory (11, 97.4)	Harmonics (36, 96.5)	Fluctuations (63, 90.5)
		Notching (37,94.5)	Fluctuations (64, 90.1)
		Fluctuations (38,95.8)	--
	Harmonics (12,97.6)	Harmonics (39,98.4)	--
	Notching (13, 97.9)	Fluctuations (40, 89.9)	--
	Fluctuations (14,96.8)	--	--
Sag (3,97.2)	Impulsive (15,94.9)	Harmonics (41,98.1)	Fluctuations (65,95.2)
		Notching (42,97.5)	Fluctuations (66,96.1)
		Fluctuations (43,97.9)	--
	Oscillatory (16,93.8)	Harmonics (44,99.1)	Fluctuations (67,98.8)
		Notching (45,90.1)	Fluctuations (68,87.8)
		Fluctuations (46,95.6)	--
	Harmonics (17,94.7)	Fluctuations (47,93.4)	--
	Notching (18,91.6)	Fluctuations (48,96.3)	--
	Fluctuations (19,96.5)	--	--
Interruption (4, 97.9)	Impulsive (20,94.7)	Harmonics (49,98.6)	Fluctuations (69,97.9)
		Notching (50,96.4)	Fluctuations (70,92.1)
	Oscillatory (21,95.4)	Fluctuations (51,97.3)	--
		Harmonics (52,91.4)	Fluctuations (71, 90.5)
		Notching (53,90.1)	Fluctuations (72,89.7)
		Fluctuations (54,91.2)	--
	Harmonics (22,96.7)	Fluctuations (55,96.2)	--
	Notching (23,91.7)	Fluctuations (56,88.7)	--
	Fluctuations (24,92.6)	--	--

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 02:08:05 UTC from IEEE Xplore. Restrictions apply.

TABLE V
(Continued.) DETAILED CLASSIFICATION ACCURACY OF THE PROPOSED MODEL UNDER 20-dB WHITE NOISE

Impulsive (5,96.8)	Harmonics (25,97.0)	Fluctuations (57,99.5)	--
	Notching (26,96.6)	Fluctuations (58,92.1)	--
	Fluctuations (27,94.5)		
Oscillatory (6,94.4)	Harmonics (28,90.3)	Fluctuations (59,87.9)	--
	Notching (29,92.1)	Fluctuations (60,89.4)	--
	Fluctuations (30,94.2)	--	--
Harmonics (7,95.9)	Fluctuations (31,91.7)	--	--
Notching (8,94.7)	Fluctuations (32,92.5)	--	--
Fluctuations (9,95.4)	--	--	--

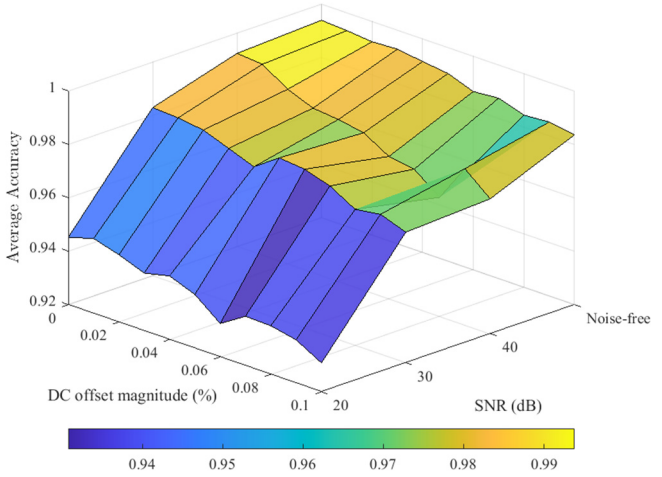


Fig. 9. Average accuracy under different levels of dc offset.

in the 40-dB white noise environment. In this environment, the average accuracy of the model decreases by 2.78% when the dc offset magnitude is 0.8%, compared to no dc offset. However, in the other noise environments, the decrease in average accuracy is controlled within 1.50%, indicating a minimal impact. DC offset has a significant impact on algorithms that rely on the peak values for PQD recognition, as it causes a shift in the peak-to-peak values of the voltage waveform. In the proposed model, which considers three characteristics of the voltage signal, the temporal–spatial feature is the most affected by dc offset, while the other two kinds of features are only slightly affected. As a result, the proposed model demonstrates strong robustness in the presence of dc offset.

The influence of frequency variations also needs to be considered, which is typically caused by an imbalance in the supply and demand of power. IEEE STD 1159-2019 defines the typical range of frequency variations as ±0.10 Hz. In the robustness test, the fundamental frequency is varied

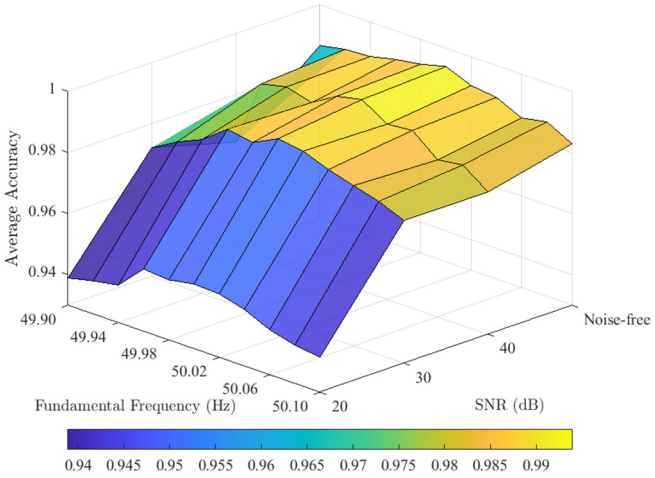


Fig. 10. Average accuracy under different levels of frequency variations.

from 49.90 to 50.10 Hz in steps of 0.02 Hz. In each noise environment, there are 500 data for each PQD type under different frequency variations. The testing results of the proposal under different noise environments and frequency variations are shown in Fig. 10.

Upon examining the surface fluctuations in Fig. 10 along the x-axis, it can be observed that the average accuracy of the proposed model is indeed affected by frequency variations. The lowest accuracy of the proposed model is 93.79%, occurring in a 20-dB white noise environment. Across four sampling environments, the accuracy decrease caused by frequency variations is kept within 2.5%. This is because a 0.1-Hz frequency change has a minimal impact on the temporal features but mostly affects the spectral features. As the proposed model comprehensively utilizes three features to identify PQD types, it exhibits strong robustness against frequency variations.

When faults or asymmetric load changes occur in the power grid, voltage phase jumps may result and affect the performance of PQD classification methods. As typical parameter descriptions for phase jumps are not found in IEEE STD 1159-2019 or other relevant standards, the robustness test is conducted based on a reference magnitude of phase jump used in [28] and [29], setting the magnitude to 30°. In the test data, the magnitude of the phase jump varies from 0° to 30° in steps of 5°, and only one time occurs randomly in each data. A total of 500 data are tested for each PQD type under the influence of phase jumps, and four sampling environments are also considered. The test results of the proposal under the influence of phase jumps in different noise environments are shown in Fig. 11.

As can be seen from the range of surface fluctuations in Fig. 11, phase jumps have a more significant effect on the performance compared to the two previous interferences. As the noise intensity and phase jump magnitude increase, the proposed model suffers more, especially in the 20-dB white noise environment, where the accuracy decreases by 4.63%. Further analysis of the misclassified samples reveals that errors mostly occur in the classification of PQDs associated with low-amplitude harmonics. This is because the

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 02:08:05 UTC from IEEE Xplore. Restrictions apply.

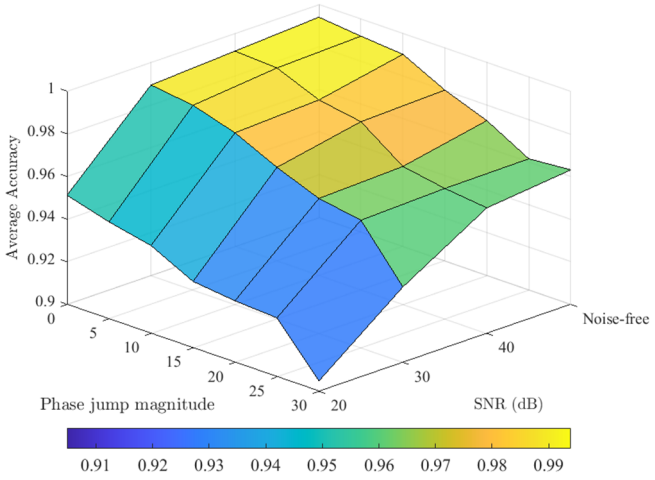


Fig. 11. Average accuracy under different levels of phase jumps.

TABLE VI
COMPARISON OF CLASSIFICATION ACCURACY WITH TRADITIONAL METHODS

Methods	Number of features	Number of PQDs	Proportion of complex PQDs	SNR (dB)	Accuracy (%)
TQWT+MSVM [13]	5	14	0.43	20	96.42
EWT+SVM [24]	6	15	0.44	25-55	95.56
WT+SVM [25]	2	10	0.60	20	94.22
ICEEMDAN +AdaKNNEO [14]	8	20	0.55	30-50	96.10
DWT+PNN [26]	9	16	0.38	20	93.60
ST+NSGA-II [27]	26	14	0.43	20	96.43
Proposal	auto	71	0.89	20	94.74

frequency-domain spatial features are the most affected by phase jumps among the three types of features. Specifically, when strong noise and significant phase jumps occur simultaneously, lower amplitude frequency components can become overwhelmed by the spectrum leakage caused by the phase jumps, leading to misjudgment. Considering the overall classification performance, the proposed model maintains an accuracy rate of over 95% in most cases, demonstrating a high level of robustness. The most significant impact of phase jumps on the model is the leakage phenomenon that occurs in the frequency domain. To further enhance the model's robustness to phase jumps, a cosine combination window can be incorporated into the FFT process to suppress spectrum leakage.

D. Performance Comparison With Traditional Algorithms

To further demonstrate the advantages of the proposed method, traditional PQDs classification methods are used as comparisons. The comparison results are analyzed in terms of features, number of PQDs, proportion of complex PQDs, and accuracy, as shown in Table VI.

The comparison results show that the number of features required by traditional methods varies, which is mainly determined by human experience, and does not constitute a unified standard. The completeness of feature expression information directly affects the classification accuracy of traditional methods. The features of different scales in the time and frequency domain are considered in "ICEEMDAN + AdaKNNEO" and "ST + NSGA-II," which can cover more PQDs types and maintain high classification accuracy, but the types of PQDs that these traditional algorithms can identify are limited due to subjective factors. Instead, the proposed method can remedy this shortcoming by automatically obtaining valid features without human help. Although the accuracy of the proposed method is not the top one, the number of PQDs and the proportion of complex PQDs are the highest, which is more in line with practical engineering applications.

TABLE VII
COMPUTATION COMPARISONS OF DIFFERENT MODELS

Models	Average training time /per epoch (s)	Total epoch	Average test time /per data (ms)
DCNN	323.63	45	119.28
SGRN	399.64	35	144.31
1D-MIR	223.58	55	80.84
Proposal	476.11	27	149.12

E. Analysis of Computation Burden

Computation is another key for applications. The training and testing time of the proposed method are recorded from Section IV-B and shown in Table VII as mean values. The single test time includes signal preprocessing (FFT and cycle segmentation) and model calculation. Meanwhile, the state-of-the-art models in Section IV-B are treated as comparisons as well.

First, considering the offline computation of each model, the proposed model has the highest average training time per epoch due to the fact that it consists of three feature extraction modules, and the BiGRU in the proposed model cannot fully take advantage of graphics processing unit (GPU) acceleration. However, in terms of the total number of epochs, the proposed model has the highest learning efficiency, taking only 27 epochs to complete training, which is eight epochs ahead of the second-ranked model. The total training duration of DCNN, SGRN, 1-D-MIR, and the proposed model is 244, 231, 207, and 212 min, respectively; thus, the proposed model is advantageous in terms of total offline training time. Then, for the single test time of each model, the FFT computation of the frequency domain input is included in the test duration of the proposed model, which ranks last among these models. However, its single test time is less than the length of a time window (200 ms), which does not cause data congestion at all and meets the requirements of online identification. At this stage, the proposed model is not fully optimizing and accelerating the model to the optimal state. In the future, it can be accelerated through strategies, such as structure optimization, model pruning, and quantization to further improve real-time performance.

F. Experiment With a Programmable AC Power Source

To verify the real-time effectiveness of the proposed model, an experimental platform is constructed, as shown in Fig. 12,

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 02:08:05 UTC from IEEE Xplore. Restrictions apply.

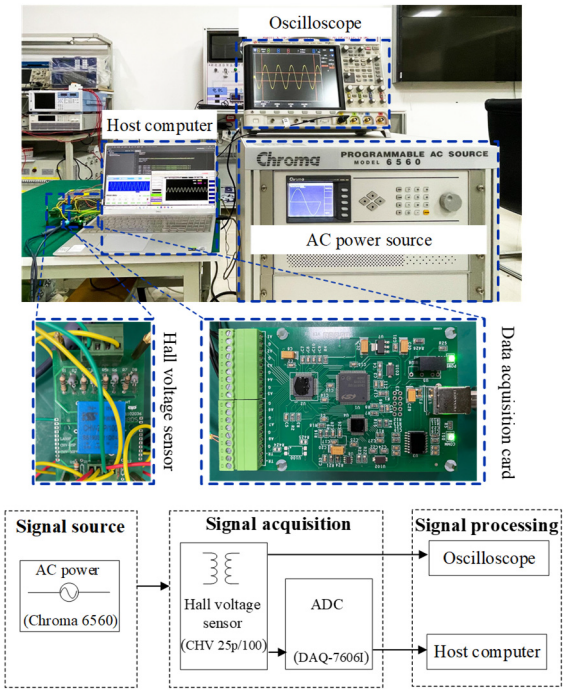


Fig. 12. Framework of the experimental platform.

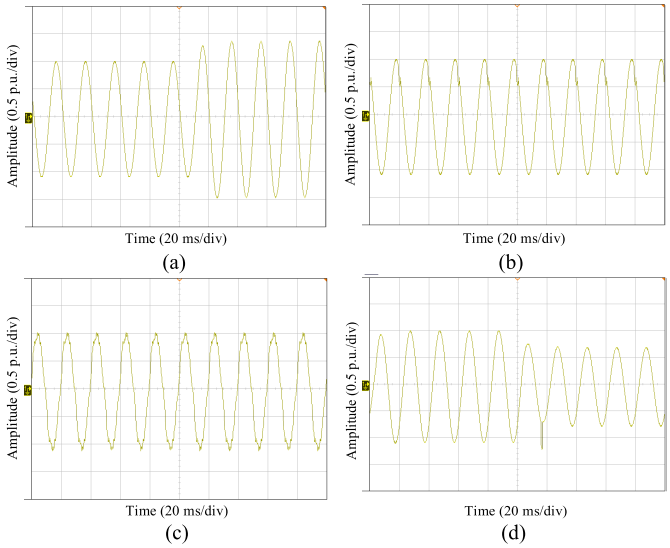


Fig. 13. PQDs waveforms captured by the oscilloscope. (a) Swell. (b) Notching. (c) Harmonics. (d) Sag + impulsive.

TABLE VIII
COMPARISON OF EXPERIMENTAL RESULTS AMONG DIFFERENT MODELS

Types of PQDs	Classification performance (%)			
	DCNN	1D-MIR	SGRN	Proposal
Swell	100	100	100	100
Sag	95	95	100	100
Interruption	100	100	90	100
Impulsive	95	90	90	95
Harmonics	100	95	100	100
Notching	100	100	80	100
Fluctuations	90	85	85	90
Swell + Harmonics	95	90	85	100
Sag + Harmonics	90	100	90	100
Swell +Notching	95	90	90	95
Sag+ Notching	100	100	95	100
Swell+ Impulsive	85	95	95	100
Sag+ Impulsive	95	95	85	95
Average accuracy	95.38	95.00	91.15	98.08

which can provide real voltage signals. The platform consists of three main parts, which are the source, the signal acquisition block, and the signal processing block. First, the signal source is implemented through the programmable ac power source of Chroma 6560, which can simulate realistic ac power supply states and waveform distortions. The output is then converted to a signal within a tolerable sampling range by a Hall voltage sensor of closed-loop hall voltage (CHV) 25p/100. Using the data acquisition card data acquisition (DAQ)-7606I, sampling can be performed at the same sampling frequency as the simulation, with an analog to digital (AD) resolution of 16 bits and a full-scale error of 0.05%. Finally, the sampled data are transmitted to the host computer through a moving window and are instantaneously classified using the proposed model. In addition, an oscilloscope is added after the Hall voltage sensor to observe the output signal of the source in real time.

The four waveforms captured by the oscilloscope are selected as examples to demonstrate the voltage waveforms containing PQD generated by the ac power supply, which are shown in Fig. 13. It can be seen that the PQDs output by the ac power source is inevitably mixed with noise.

Due to hardware limitations of the ac power supply, only a limited range of single and multiple PQD can be generated. The signals tested in the experiment consist of seven single PQDs and six multiple PQDs. A total of 50 data are generated for each type, with their disturbance parameters randomly set. Ten cycles are sampled at a sampling frequency of 3200 Hz. The final classification test results of the proposed model are shown in Table VIII. Moreover, classification results based on other deep learning models are presented for comparison.

Comparing the classification results of each model on the experimental signals, it can be seen that the proposed model is still leading in accuracy, outperforming DCNN, 1-D-MIR, and SGRN by 2.70%, 3.08%, and 6.93%, respectively. In addition, the proposed algorithm has obvious advantages for the classification of both single and complex PQDs. Among the 13 PQDs, the proposed model has a relatively poor classification performance on fluctuations. Upon further analysis of the misclassified fluctuation data, it is found that all of them exhibited low-frequency and low-amplitude characteristics, which makes them prone to being misidentified as normal signals. For example, one sample has a frequency and amplitude of 5 Hz and 4%, respectively. The low-frequency fluctuations are difficult to detect within a 10-cycle window. However, disturbances with parameters close to the lower limit values would not have a significant impact on the power grid. Additionally, the proposed model correctly identified the remaining 48 fluctuation data.

To test the robustness of the model in an experimental setting, two additional test cases are implemented, which consider the effects of frequency variation and phase jumps, respectively. Due to hardware limitations, it is currently not possible to test the effect of dc offset. In these cases, the interference occurrence time is randomly set, and the frequency

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 02:08:05 UTC from IEEE Xplore. Restrictions apply.

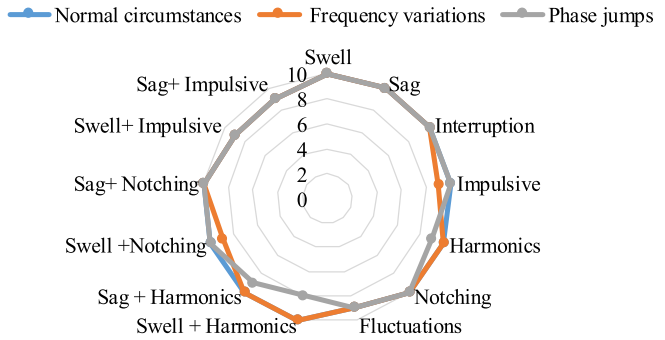


Fig. 14. Impact of frequency variations and phase jumps on the proposed model's experimental performance.

and phase changes are set at 0.10 Hz and 30°, respectively. For different cases, each PQD contains ten sets to test the model. Finally, the test results are presented in the form of a radar chart, as shown in Fig. 14.

The radar curves in Fig. 14 represent the number of correctly recognized PQD samples by the proposed model. First, comparing the orange and blue curves reveals that frequency variations have a negligible impact on the model's performance, with only two more misclassified samples compared to the normal circumstances. Second, contrasting the gray and blue curves shows that phase jumps have a consistent effect on the model's recognition rate as observed in simulations, meaning that they can easily lead to misclassification of PQD samples with harmonics, resulting in four more misclassified samples compared to the normal circumstances. The level of overlap between the three curves indicates that the proposed model also exhibits strong robustness in experimental tests when frequency variations and phase jumps are considered.

Overall, the above presented results demonstrate that the proposed model can efficiently identify test signals in the experimental platform, further validating its reliability.

V. Conclusion

In this article, a multidimensional feature-driven ensemble model is proposed for the classification task of 71 PQDs, including eight single PQDs and 63 complex PQDs. Compared to existing deep learning-based methods, the proposed model considers a more comprehensive range of features, including spatial features in the time–frequency domain and temporal relational features. According to the characteristics of different dimensions, the corresponding adaptor modules are designed based on FCN and BiGRU networks, and the channel attention mechanism of SENet is introduced to further improve the training efficiency and recognition accuracy. The simulation results unequivocally demonstrate the superiority of the proposed model over other deep learning models, both in terms of training efficiency and classification accuracy. Robustness tests further reveal the reliability of the model, as it exhibits minimal fluctuations when exposed to external factors. Of particular note is the construction of a hardware platform that utilizes an ac power source to generate realistic PQDs, while considering interference conditions. This development effectively establishes the practicality of the proposed model in engineering applications. However, as the proposed model demonstrates a significant improvement in classification performance, the computational cost for both training and testing inevitably increases. Therefore, future research will focus on exploring ways to optimize the software and hardware of the model to improve its efficiency and running speed.

References

[1] H. Husin and M. Zaki, "A critical review of the integration of renewable energy sources with various technologies," *Protection Control Modern Power Syst.*, vol. 6, no. 1, Dec. 2021, Art. no. 116615, doi: 10.1186/s41601-021-00181-3.

[2] M. A. Mohamed, T. Jin, and W. Su, "Multi-agent energy management of smart islands using primal-dual method of multipliers," *Energy*, vol. 208, Oct. 2020, Art. no. 118306, doi: 10.1016/j.energy.2020.118306.

[3] S. Elphick, P. Ciufo, G. Drury, V. Smith, S. Perera, and V. Gosbell, "Large scale proactive power-quality monitoring: An example from Australia," *IEEE Trans. Power Del.*, vol. 32, no. 2, pp. 881–889, Apr. 2017, doi: 10.1109/TPWRD.2016.2562680.

[4] Y. Xu, S. Fan, S. Xie, and M. Lu, "Power quality detection and classification in high permeability active distribution network based on IEWT-MFDE," *CSEE J. Power Energy Syst.*, vol. 8, no. 6, pp. 1–13, 2022, doi: 10.17775/CSEEJPES.2020.00110.

[5] I. Parvez, M. Aghili, A. I. Sarwat, S. Rahman, and F. Alam, "Online power quality disturbance detection by support vector machine in smart meter," *J. Modern Power Syst. Clean Energy*, vol. 7, no. 5, pp. 1328–1339, Sep. 2019, doi: 10.1007/s40565-018-0488-z.

[6] W. Zhao, L. Shang, and J. Sun, "Power quality disturbance classification based on time-frequency domain multi-feature and decision tree," *Protection Control Modern Power Syst.*, vol. 4, no. 1, p. 183, Dec. 2019, doi: 10.1186/s41601-019-0139-z.

[7] Z. Liu, Y. Cui, and W. Li, "A classification method for complex power quality disturbances using EEMD and rank wavelet SVM," *IEEE Trans. Smart Grid*, vol. 6, no. 4, pp. 1678–1685, Jul. 2015, doi: 10.1109/TSG.2015.2397431.

[8] Y. Xu, Y. Gao, Z. Li, and M. Lu, "Detection and classification of power quality disturbances in distribution networks based on VMD and DFA," *CSEE J. Power Energy Syst.*, vol. 6, no. 1, pp. 122–130, 2020, doi: 10.17775/CSEEJPES.2018.01340.

[9] P. D. Achlerkar, S. R. Samantaray, and M. S. Manikandan, "Variational mode decomposition and decision tree based detection and classification of power quality disturbances in grid-connected distributed generation system," *IEEE Trans. Smart Grid*, vol. 9, no. 4, pp. 3122–3132, Jul. 2018, doi: 10.1109/TSG.2016.2626469.

[10] S. Karasu and Z. Saraç, "Classification of power quality disturbances by 2D-Riesz transform, multi-objective grey wolf optimizer and machine learning methods," *Digit. Signal Process.*, vol. 101, Jun. 2020, Art. no. 102711, doi: 10.1016/j.dsp.2020.102711.

[11] M. V. Reddy and R. Sodhi, "A modified S-transform and random forests-based power quality assessment framework," *IEEE Trans. Instrum. Meas.*, vol. 67, no. 1, pp. 78–89, Jan. 2018, doi: 10.1109/TIM.2017.2761239.

[12] J. Li, Z. Teng, Q. Tang, and J. Song, "Detection and classification of power quality disturbances using double resolution S-Transform and DAG-SVMs," *IEEE Trans. Instrum. Meas.*, vol. 65, no. 10, pp. 2302–2312, Oct. 2016, doi: 10.1109/TIM.2016.2578518.

[13] K. Thirumala, M. S. Prasad, T. Jain, and A. C. Umarikar, "Tunable-Q wavelet transform and dual multiclass SVM for online automatic detection of power quality disturbances," *IEEE Trans. Smart Grid*, vol. 9, no. 4, pp. 3018–3028, Jul. 2018, doi: 10.1109/TSG.2016.2624313.

[14] Y. Liu, T. Jin, M. A. Mohamed, and Q. Wang, "A novel three-step classification approach based on time-dependent spectral features for complex power quality disturbances," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–14, 2021, doi: 10.1109/TIM.2021.3050187.

[15] L. Lin, D. Wang, S. Zhao, L. Chen, and N. Huang, "Power quality disturbance feature selection and pattern recognition based on image enhancement techniques," *IEEE Access*, vol. 7, pp. 67889–67904, 2019, doi: 10.1109/ACCESS.2019.2917886.

[16] K. Cai, W. Cao, L. Aarniovuori, H. Pang, Y. Lin, and G. Li, "Classification of power quality disturbances using wigner-ville distribution and deep convolutional neural networks," *IEEE Access*, vol. 7, pp. 119099–119109, 2019, doi: 10.1109/ACCESS.2019.2937193.

[17] J. Shukla, B. K. Panigrahi, and P. K. Ray, "Power quality disturbances classification based on gramian angular summation field method and convolutional neural networks," *Int. Trans. Electr. Energy Syst.*, vol. 31, no. 12, p. 1748, Dec. 2021, doi: 10.1002/2050-7038.13222.

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 02:08:05 UTC from IEEE Xplore. Restrictions apply.

[18] S. Wang and H. Chen, "A novel deep learning method for the classification of power quality disturbances using deep convolutional neural network," *Appl. Energy*, vol. 235, pp. 1126–1140, Feb. 2019, doi: 10.1016/j.apenergy.2018.09.160.

[19] R. Gong and T. Ruan, "A new convolutional network structure for power quality disturbance identification and classification in micro-grids," *IEEE Access*, vol. 8, pp. 88801–88814, 2020, doi: 10.1109/ACCESS.2020.2993202.

[20] X. Zu and K. Wei, "A simple gated recurrent network for detection of power quality disturbances," *IET Gener., Transmiss. Distribution*, vol. 15, no. 4, pp. 751–761, Feb. 2021, doi: 10.1049/gtd2.12056.

[21] L. Ortiz, R. Orizondo, A. Águila, J. W. González, G. J. López, and I. Isaac, "Hybrid AC/DC microgrid test system simulation: Grid-connected mode," *Heliyon*, vol. 5, no. 12, Dec. 2019, Art. no. e02862, doi: 10.1016/j.heliyon.2019.e02862.

[22] A. G. Shaik and O. P. Mahela, "Power quality assessment and event detection in hybrid power system," *Electric Power Syst. Res.*, vol. 161, pp. 26–44, Aug. 2018, doi: 10.1016/j.epsr.2018.03.026.

[23] C. Zhao et al., "Novel method based on variational mode decomposition and a random discriminative projection extreme learning machine for multiple power quality disturbance recognition," *IEEE Trans. Ind. Informat.*, vol. 15, no. 5, pp. 2915–2926, May 2019, doi: 10.1109/TII.2018.2871253.

[24] K. Thirumala, S. Pal, T. Jain, and A. C. Umarikar, "A classification method for multiple power quality disturbances using EWT based adaptive filtering and multiclass SVM," *Neurocomputing*, vol. 334, pp. 265–274, Mar. 2019, doi: 10.1016/j.neucom.2019.01.038.

[25] M. D. Borras, J. C. Bravo, and J. C. Montano, "Disturbance ratio for optimal multi-event classification in power distribution networks," *IEEE Trans. Ind. Electron.*, vol. 63, no. 5, pp. 3117–3124, May 2016, doi: 10.1109/TIE.2016.2521615.

[26] S. Khokhar, A. A. M. Zin, A. P. Memon, and A. S. Mokhtar, "A new optimal feature selection algorithm for classification of power quality disturbances using discrete wavelet transform and probabilistic neural network," *Measurement*, vol. 95, pp. 246–259, Jan. 2017, doi: 10.1016/j.measurement.2016.10.013.

[27] U. Singh and S. N. Singh, "Optimal feature selection via NSGA-II for power quality disturbances classification," *IEEE Trans. Ind. Informat.*, vol. 14, no. 7, pp. 2994–3002, Jul. 2018, doi: 10.1109/TII.2017.2773475.

[28] Y. Xi, Z. Li, X. Zeng, and X. Tang, "Detection of power quality disturbances using an adaptive process noise covariance Kalman filter," *Digit. Signal Process.*, vol. 76, pp. 34–49, May 2018, doi: 10.1016/j.dsp.2018.01.013.

[29] C. Liang et al., "Improved S-transform for time-frequency analysis for power quality disturbances," *IEEE Trans. Power Del.*, vol. 37, no. 4, pp. 2942–2952, Aug. 2022, doi: 10.1109/TPWRD.2021.3119918.



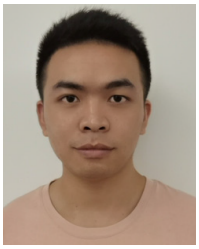
Yulong Liu received the B.S. and M.S. degrees from China Three Gorges University, Yichang, China, in 2014 and 2017, respectively. He is currently pursuing the Ph.D. degree with the College of Electrical Engineering and Automation, Fuzhou University, Fuzhou, China.

His current research interest includes power quality disturbance detection and harmonics detection of power systems.



Ding Yuan received the B.S. degree from Yanshan University, Qinhuangdao, China, in 2020. He is currently pursuing the M.S. degree with the College of Electrical Engineering and Automation, Fuzhou University, Fuzhou, China.

His current research interest includes power quality disturbance detection.



Hongwei Fan received the B.S. degree from the Nanjing Institute of Engineering, Nanjing, China, in 2019. He is currently pursuing the M.S. degree with the College of Electrical Engineering and Automation, Fuzhou University, Fuzhou, China.

His current research interest includes energy router research.



Tao Jin (Senior Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from Yanshan University, Qinhuangdao, China, in 1998 and 2001, respectively, and the Ph.D. degree in electrical engineering from Shanghai Jiaotong University, Shanghai, China, in 2005.

From 2005 to 2007, he worked as a Post-Doctoral with Shanghai Jiaotong University. From 2008 to 2009, he held a Research Scientist Position with Virginia Tech, Blacksburg, VA, USA. In 2010, he joined Imperial College London, London, U.K., as European Union Marie Curie Research Fellow, where he focused on electrical technologies related to smart grid. He is currently a Professor with the College of Electrical Engineering and Automation, Fuzhou University, Fuzhou, China. He has authored about 200 articles.

Dr. Jin is a member of the IEEE Power and Energy Society and the IEEE Industrial Electronics Society, and a Special Committee Member of the Chinese Society of Electrical Engineering and the China Electrotechnical Society. He currently serves as Co-Editor-in-Chief of ESTA, and an Associate Editors for MPCE, PCMP, High Voltage, China Measurement and Testing Technology, and other journals.



Mohamed A. Mohamed (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical engineering from Minia University, Minia, Egypt, in 2006 and 2010, respectively, and the Ph.D. degree in electrical engineering from King Saud University, Riyadh, Saudi Arabia, in 2016.

He joined the College of Electrical Engineering and Automation, Fuzhou University, Fuzhou, China, as a Post-Doctoral Research Fellow, in 2018. He has been a Faculty Member with the Department of Electrical Engineering, College of Engineering, Minia University, since 2008. His research interests include power system analysis, renewable energy integration, energy management, power electronics, electrical vehicles, optimization, smart islands, smart cities, and smart grids. He has supervised multiple M.Sc. and Ph.D. theses, worked on a number of technical projects, and authored various articles and books.

Dr. Mohamed has also joined the editorial board of several scientific journals and the steering committees of many international conferences.

Authorized licensed use limited to: VIT University. Downloaded on August 21,2026 at 02:08:05 UTC from IEEE Xplore. Restrictions apply.